Fiscal Year:	FY 2010	**Task Last Updated:**	FY 08/15/2010
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Manipulation of two-phase flow in low-gravity environments: drop-phase relocation by active wettability control		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/31/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Duval, W.	**Contact Phone:**	216-433-5023
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Daniel, Susan (Cornell University) Hui, Chung Yuen (Cornell University)		
Grant/Contract No.:	NNX09AI83G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Development of a means to electronically control wettability of a substrate surface in real time opens an opportunity to manipulate two-phase flow in the absence of earth's gravity. Condensers and separators are a crucial part of life support and power systems. It is proposed to develop a proof-of-concept device for collecting drops on a surface in low-gravity. A no-gravity cyclone separator would serve as a practical proof-of-concept of the collection technology. Motivation comes from an earth-based observation that passive chemical gradients can propel condensing stream droplets to a collector with centimeter per second speeds. Note that thermal effects are thought not to be central to the mechanism. What is needed for propelling drops is a bias, provided by the chemical gradient, and fluctuations, sufficient to overcome the contact-line tendency to pin. It is proposed to develop a tunable gradient to serve as a controllable bias and to develop a local vibration source to serve as the fluctuations. The goal is the capability to move drops on-demand in a number of directions.		

	A scientific basis requires understanding the nature of the contact-line, contact-angle and spreading behavior, all of which can be influenced by the global drop dynamics. Computations are therefore also proposed to predict which drop oscillations will be most effective in de-pinning the contact-line and which vibrations will be most effective. Detachment will also be examined both experimentally and computationally since understanding how droplets get entrained in the bulk flow depends on the liquid/solid surface energy which will be tunable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The location of the liquid- and gas-phases in two-phase liquid/gas flows on Earth depends crucially on the strength of Earth's gravity. Many devices that perform well on Earth are unreliable for lunar and Mars missions. They falter in low-gravity. For example, the Total Organic Carbon [1] analyzer that works well on Earth is ‘currently not in use due to difficulty in bubble removal.’ Hence, monitoring the quality of on-board water supplies becomes a high risk (red-light) issue for lunar and Mars missions (risk 38 of the Bioastronautic Critical Path Roadmap (BCPR) [1]). This is not an isolated instance. Condensers are crucial to Rankine-cycle power plants [2], among other applications, and for regenerative Life Support systems (CO2 condenser/separators) [3]. Additionally, success of this research effort will likely lead to better earth-based two-phase flow systems, including condensers and other heat-exchange devices. 1. Center, J. S., Bioastronautic Critical Path Roadmap, Supporting Documents. 2. Lahey, R. T.; Dhir, V., Research in Support of the Use of Rankine Cycle Energy Conversion Systems for Space Power and Propulsion. NASA/CR-2004-213142 2004. 3. [NAP2000], Microgravity Research in Support of Technologies for the Human Exploration and Development of Space and Planetary Bodies. National Academy Press: 2000.
Task Progress:	The scientific goal of this project is to move droplets in a directed fashion using passive and active surface-energy gradients. The technological goal is to control two-phase flow in low-gravity using active wettability. Progress toward scientific and technological goals is presented. On the side of science, the study of capillary vibrations of constrained inviscid liquids began with the cylindrical interface, where the computational implementation of the linearized governing equations could be tested against the literature in a number of limiting cases. The computational technique has proved to be robust. The geometry next shifted to sessile spherical-drops (undisturbed shape) subjected to one of four contact-line constitutive behaviors: i) pinned contact-line; ii) fixed contact-angle; iii) continuous contact-angle vs speed response; and iv) slip of contact-line. Solutions that deliver the frequency spectrum of sessile drops allowing for vibration-induced de-pinning of the contact-line are sought. This work has led to new solutions of the inviscid governing equations, solutions where symmetry breaking serves to organize the frequency spectrum of the drops. Experiments are in progress to confirm the predictions of these new solutions. Mechanical in-plane vibrations supply the forcing. On the side of technology, a localized means to de-pin the drops is needed. Mechanical in-plane vibrations are difficult to localize. The proposed alternative is to use individually-activated droplets covered with a thin elastic film -- not unlike a water balloon but on a sub-millimeter scale to enable localization. Activation will eventually use electroosmotic forcing. Before the active surface can be implemented, however, the mechanical behavior of the balloon must be known. The pressure-volume response for balloons made from typical nonlinear elastic materials, such as PDMS or latex rubber, is being measured.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Bostwick JB, Steen PH. "Oscillations of a viscous drop under spherical-belt constraint." Presented at the 62nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Minneapolis, MN, November 24, 2009. Bulletin of the 2009 APS/DFD Annual Meeting, November 2009. , Nov-2009
Articles in Peer-reviewed Journals	Bostwick JB, Steen PH. "Stability of constrained cylindrical interfaces and the torus lift of Plateau-Rayleigh." Journal of Fluid Mechanics 2010 Mar;647:201-19. http://dx.doi.org/10.1017/S0022112009993831 , Mar-2010